



National Textile University
Department of Computer Science

Subject

Operating System

Submitted to:

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Submitted by:

Ahmad Farooq

Registration Number

23-NTU-CS-1128

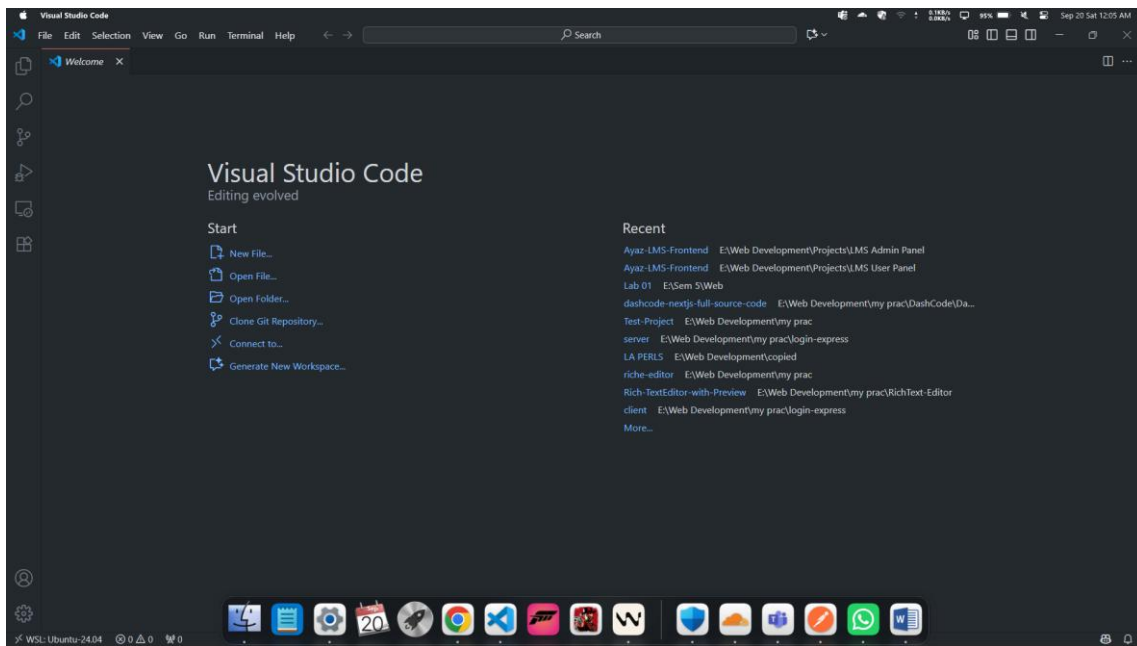
Lab No.

01

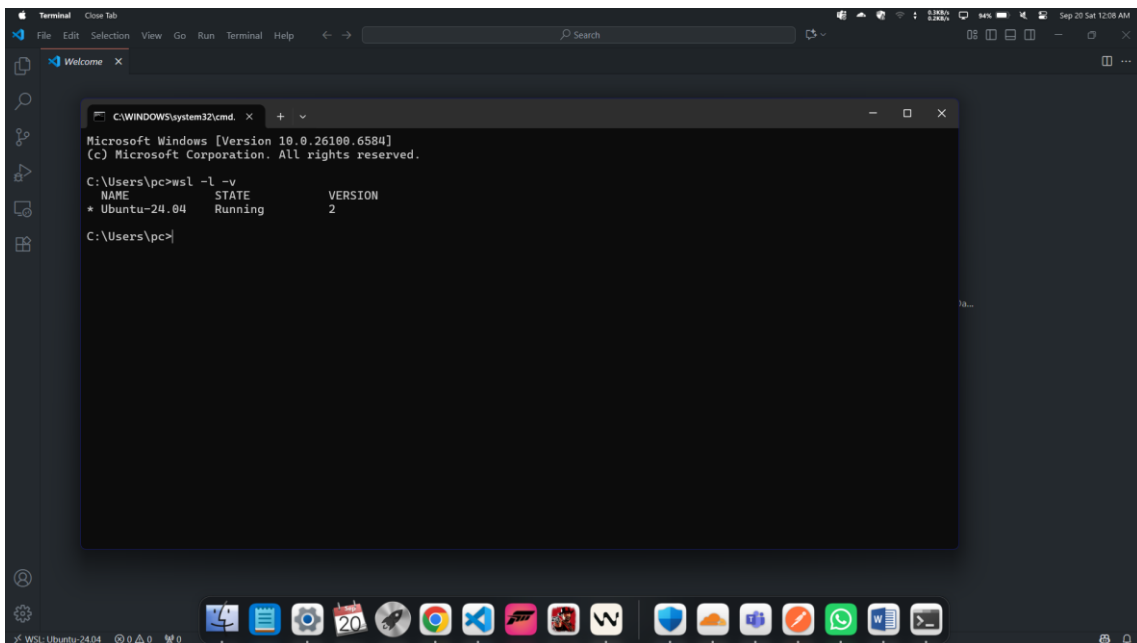
Semester

5th

Connecting VS code with WSL :



WSL version and Ubuntu runing:



APT Update:

```
Microsoft Windows [Version 10.0.26100.6584]
(c) Microsoft Corporation. All rights reserved.

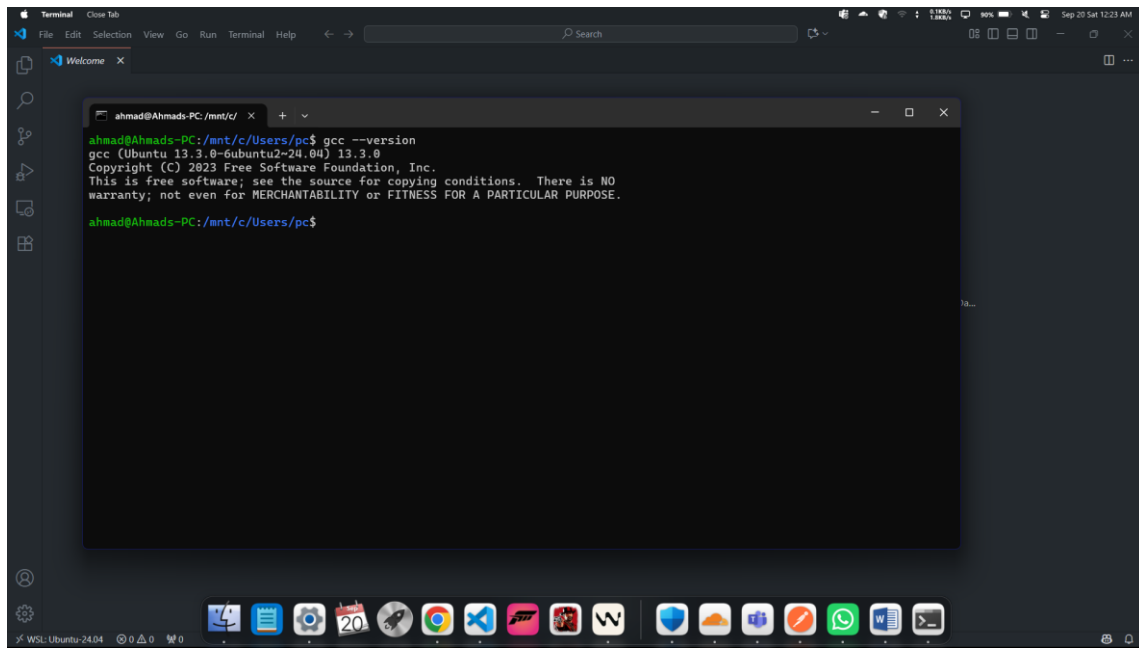
C:\Users\pc>wsl -l -v
      NAME      STATE      VERSION
* Ubuntu-24.04  Running      2

C:\Users\pc>wsl
ahmad@Ahmads-PC: /mnt/c/Users/pc$ sudo apt update
[sudo] password for ahmad:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1143 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [8744 B]
Get:8 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [880 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1415 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [195 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.2 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [18.0 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [288 B]
Get:14 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [520 B]
Get:15 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [288 B]
Get:16 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [384 B]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [278 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [15.3 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1485 kB]
```

Gcc Install:

```
ahmad@Ahmads-PC: /mnt/c/Users/pc$ sudo apt install gcc -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  cpp cpp-13 cpp-13-x86-64-linux-gnu gcc gcc-13 gcc-13-base gcc-13-x86-64-linux-gnu
  gcc-x86-64-linux-gnu libasan8 libatomic1 libc-dev-bin libc-devtools libc6-dev libcc1-0 libcrypt-dev
  libde265-0 libgcc-13-dev libgd3 libgomp1 libheif-plugin-aomdec libheif-plugin-aomenc libheif-plugin-libde265
  libheif1 libhwasan0 libisl23 libitm1 liblsan0 libmpc3 libquadmath0 libtsan2 libubsan1 libxpm4 linux-libc-dev
  manpages-dev rpcsvc-proto
Suggested packages:
  cpp-doc gcc-13-locales cpp-13-doc gcc-multilib make autoconf automake libtool flex bison gdb gcc-doc gcc-multilib
  gcc-13-doc gdb-x86-64-linux-gnu glibc-doc libgd-tools libheif-plugin-x265 libheif-plugin-ffmpegdec
  libheif-plugin-jpegdec libheif-plugin-jpegenc libheif-plugin-j2kdec libheif-plugin-j2kenc libheif-plugin-rav1e
  libheif-plugin-svtenc
The following NEW packages will be installed:
  cpp cpp-13 cpp-13-x86-64-linux-gnu gcc gcc-13 gcc-13-base gcc-13-x86-64-linux-gnu
  gcc-x86-64-linux-gnu libasan8 libasan8 libatomic1 libc-dev-bin libc-devtools libc6-dev libcc1-0 libcrypt-dev
  libde265-0 libgcc-13-dev libgd3 libgomp1 libheif-plugin-aomdec libheif-plugin-aomenc libheif-plugin-libde265
  libheif1 libhwasan0 libisl23 libitm1 liblsan0 libmpc3 libquadmath0 libtsan2 libubsan1 libxpm4 linux-libc-dev
  manpages-dev rpcsvc-proto
0 upgraded, 37 newly installed, 0 to remove and 3 not upgraded.
Need to get 55.0 MB of archives.
After this operation, 181 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gcc-13-base amd64 13.3.0-6ubuntu2-24.04 [51.5 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libisl23 amd64 0.26-3build1.1 [680 kB]
1% [2 libisl23 442 kB/680 kB 65%] 41.2 kB/s 22min 3s
```

Gcc version:



The image shows a terminal window within a desktop environment. The terminal has a title bar that reads "Terminal" and "Close Tab". Below the title bar is a menu bar with options: File, Edit, Selection, View, Go, Run, Terminal, and Help. A search bar is also present. The terminal content shows a user named "ahmad" at a prompt "ahmad@Ahmads-PC:/mnt/c/Users/pc\$". The user has entered the command "gcc --version". The output of the command is displayed in green text: "gcc (Ubuntu 13.3.0-6ubuntu2-24.04) 13.3.0", "Copyright (C) 2023 Free Software Foundation, Inc.", and "This is free software; see the source for copying conditions. There is NO warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE." The prompt "ahmad@Ahmads-PC:/mnt/c/Users/pc\$" is shown again, indicating the command has finished. The desktop environment includes a dock at the bottom with various application icons like a calendar, file manager, and web browser. The system status bar at the top right shows the date and time as "Sep 20 Sat 12:23 AM".

```
ahmad@Ahmads-PC:/mnt/c/Users/pc$ gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2-24.04) 13.3.0
Copyright (C) 2023 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
ahmad@Ahmads-PC:/mnt/c/Users/pc$
```